

Interrupt Controller

Programmable priority and preemption

REQUESTS



ARBITRATION



SERVICE

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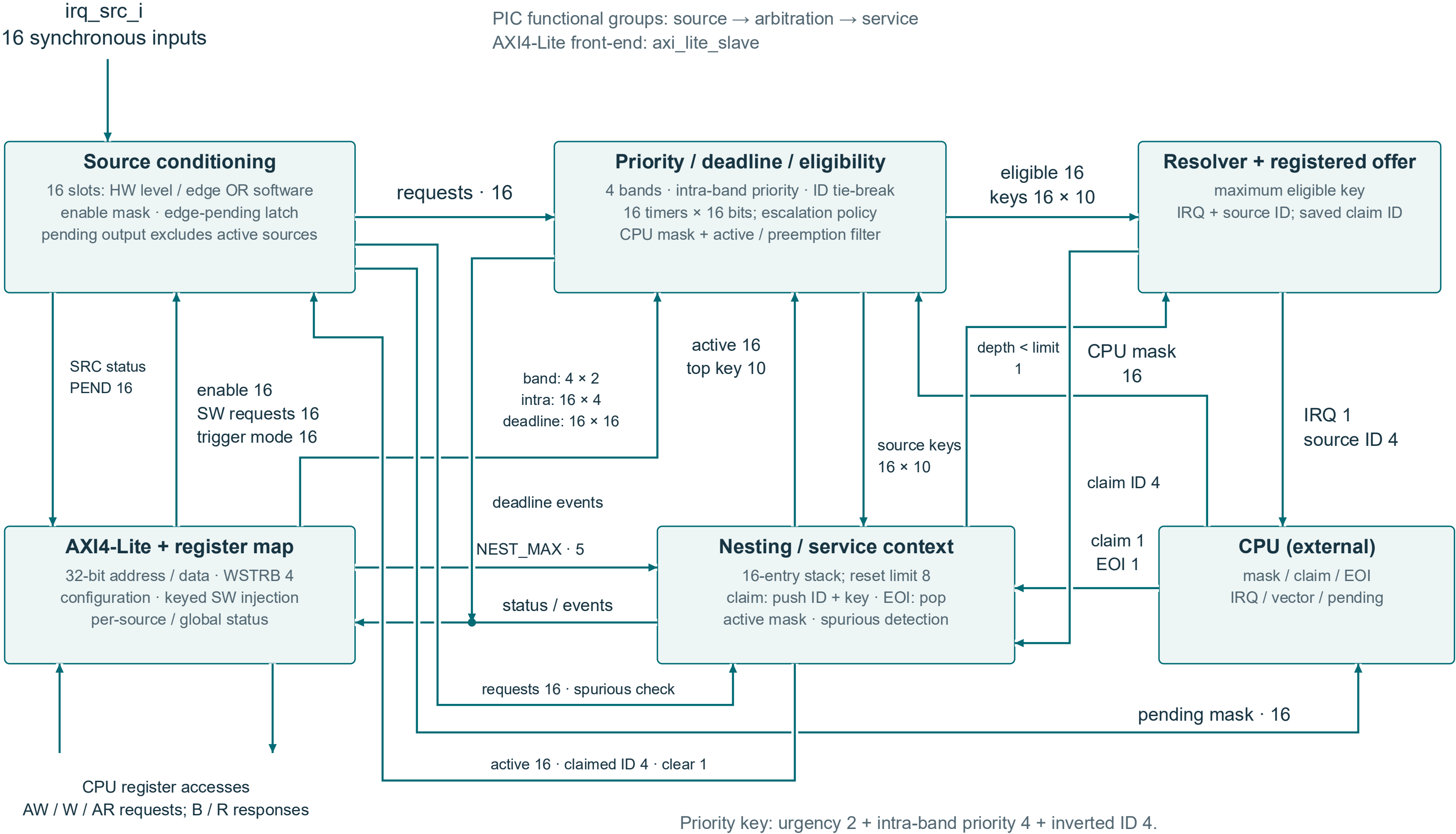
RTL implementation · interfaces · verification

The task: a PIC with priority bands, nesting and deadlines

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|-----------|---|-----------|---|
| 01 | 16 hardware sources + 16 software triggers | 06 | Detection and reporting of spurious requests |
| 02 | Level / edge selection for every source | 07 | Per-source deadline and automatic escalation |
| 03 | Priority bands and intra-band arbitration | 08 | Protected software trigger and a defined clear rule |
| 04 | Preemption that preserves the active contexts | 09 | AXI4-Lite: configuration and status registers |
| 05 | Programmable limit for the nesting depth | 10 | SW interrupts identical to HW once they enter arbitration |

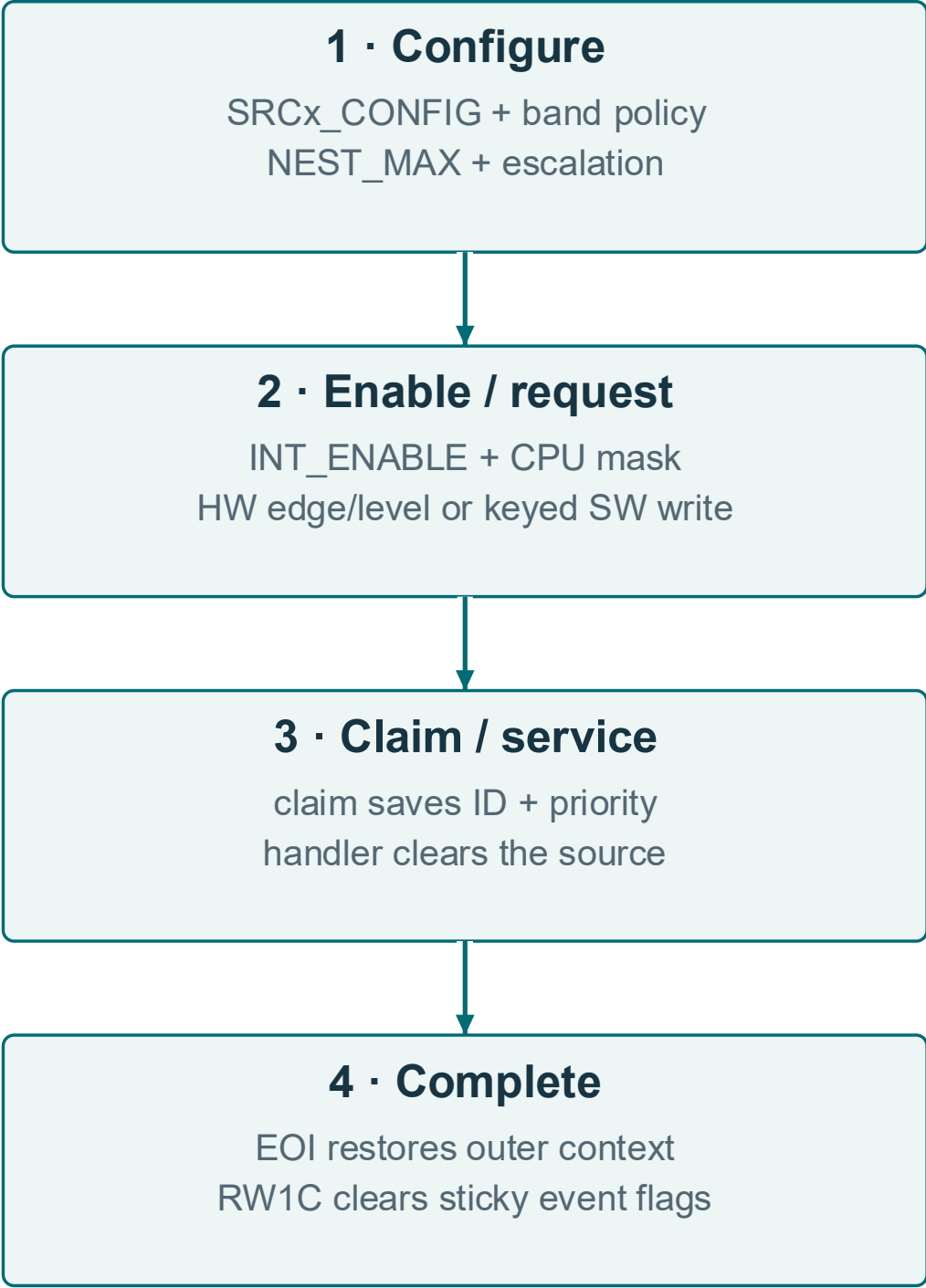
The 32 HW/SW inputs merge into 16 source identities.

The selected request becomes an active context only at claim



The registers set the policy and make the state observable

Address	Register / group	Role in use
0x00 + 4n	SRCn_CONFIG	Trigger, band, priority, deadline
0x40 + 4n	SRCn_SW_TRIG	Injection with key 0xA5A5; clear
0x80 + 4n	SRCn_STATUS	Pending, active, escalated, spurious
0xC0 / D4	BAND_CONFIG / ESCALATION_CFG	Band urgencies and escalation policy
0xC4 / C8 / CC	NEST_STATUS / NEST_MAX / ACTIVE_VEC	Active context and nesting limit
0xD8	INT_ENABLE	Per-source enable
0xD0 / DC	SPURIOUS_LOG / INT_STATUS	Logged events; cleared by RW1C

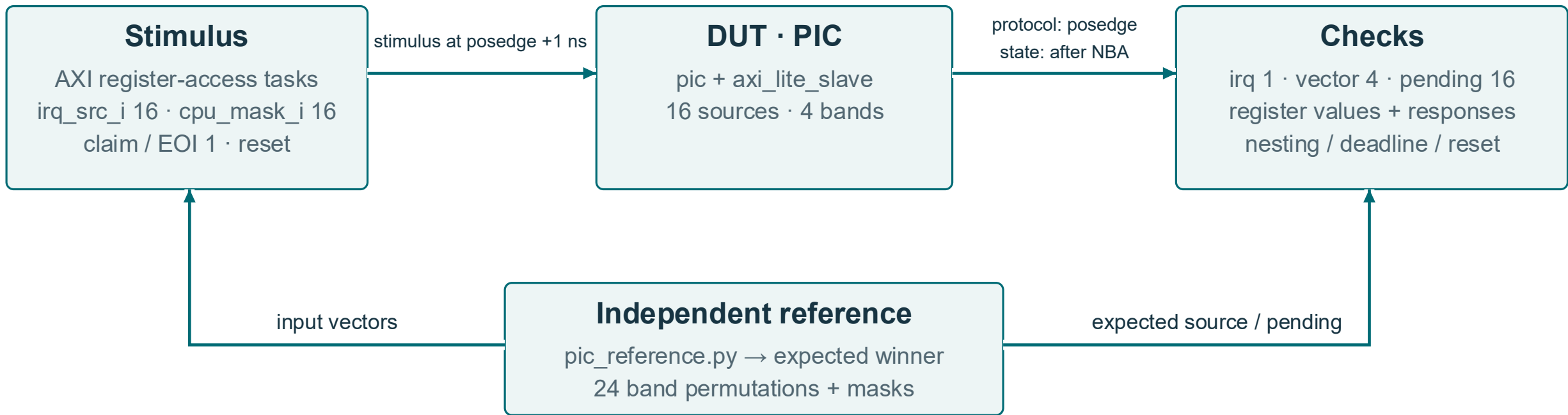


Register configuration → service sequence

The functional result and the bus protocol are checked separately

Directed scenarios plus an independent priority model

Protocol and PIC invariants are checked by assertions attached through bind.



Independent model

24 urgency permutations + masks + ties

Protocol + state

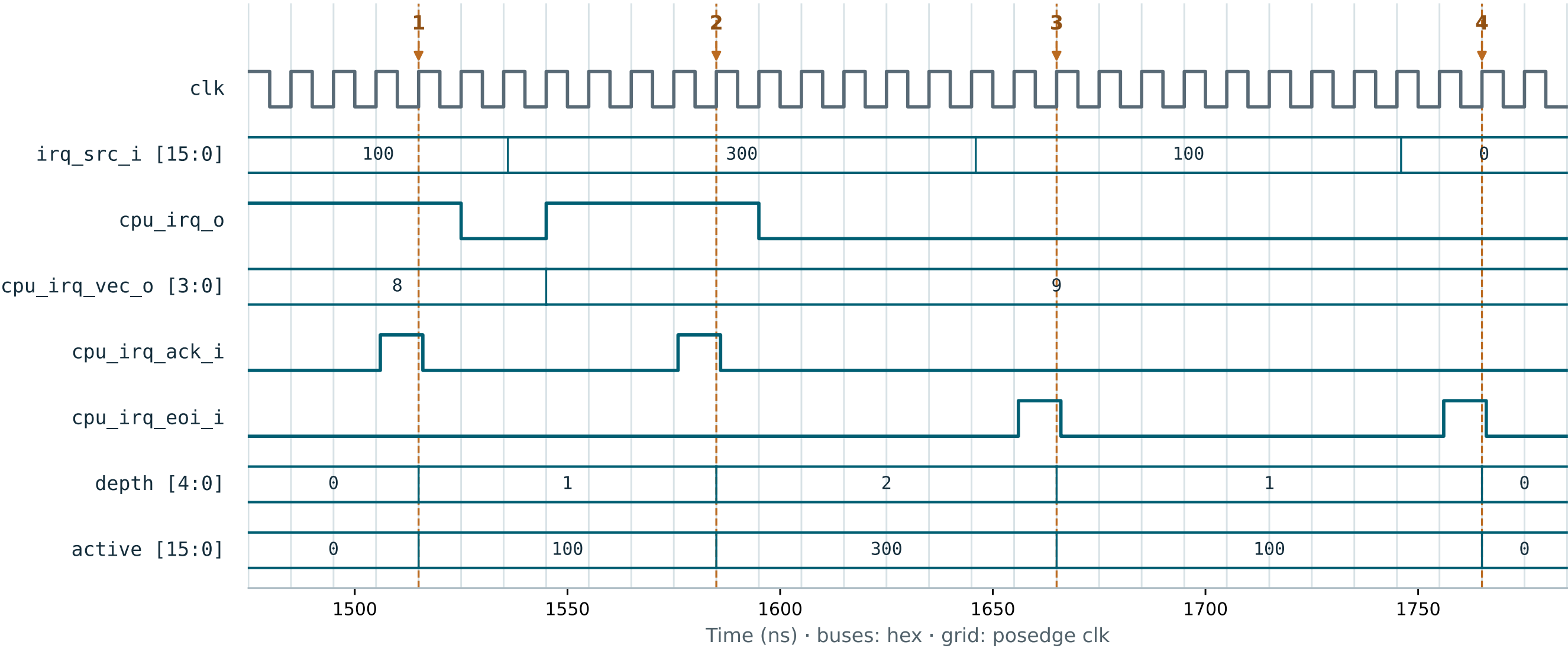
AXI monitors, registers, stack and SVA through bind

Critical cases

edge + claim · nesting · deadline · async reset

ModelSim: 22 CPU/PIC configurations · Verilator: 16 benches with assertions

Preemption keeps the context; two EOIs close it in order



1 · Claim 8

depth 1 · active 0x100

2 · Claim 9

depth 2 · active 0x300

3 · Inner EOI

depth 1 · source 8 stays active

4 · Outer EOI

depth 0 · active 0

Bus values are hexadecimal; the offered vector is valid while IRQ = 1.

Result: configurable arbitration with verified preemption

WHAT I BUILT

1. Banded priority, nesting and deadlines, all configurable
2. Registers and CPU handshake integrated with arbitration
3. Policy checked independently and in concurrent tests

WHAT I LEARNED

- I defined tie cases and the priority of simultaneous events.
- I tracked the saved context, not only the winning vector.
- I checked arbitration against a model independent of the RTL.

My contribution: design decisions → RTL → integration → verification